

d-separation

CSCI
270

independence can be tough to "see"
in a probability distribution

A	B	C	P_1
0	0	0	$\frac{1}{32}$
0	0	1	$\frac{3}{32}$
0	1	0	$\frac{6}{32}$
0	1	1	$\frac{6}{32}$
1	0	0	$\frac{3}{32}$
1	0	1	$\frac{1}{32}$
1	1	0	$\frac{4}{32}$
1	1	1	$\frac{8}{32}$

A	B	C	P_2
0	0	0	$\frac{1}{32}$
0	0	1	$\frac{3}{32}$
0	1	0	$\frac{6}{32}$
0	1	1	$\frac{6}{32}$
1	0	0	$\frac{3}{32}$
1	0	1	$\frac{3}{32}$
1	1	0	$\frac{2}{32}$
1	1	1	$\frac{8}{32}$

is $A \perp B$?

A	B	C	P_1	A	B	C	P_2
0	0	0	$\frac{1}{32}$	0	0	0	$\frac{1}{32}$
0	0	1	$\frac{3}{32}$	0	0	1	$\frac{3}{32}$
0	1	0	$\frac{6}{32}$	0	1	0	$\frac{6}{32}$
0	1	1	$\frac{6}{32}$	0	1	1	$\frac{6}{32}$
1	0	0	$\frac{3}{32}$	1	0	0	$\frac{3}{32}$
1	0	1	$\frac{1}{32}$	1	0	1	$\frac{3}{32}$
1	1	0	$\frac{4}{32}$	1	1	0	$\frac{2}{32}$
1	1	1	$\frac{8}{32}$	1	1	1	$\frac{8}{32}$

is $A \perp\!\!\!\perp B$?

A	B	C	P_i
0	0	0	$\frac{1}{32}$
0	0	1	$\frac{3}{32}$
0	1	0	$\frac{6}{32}$
0	1	1	$\frac{6}{32}$
1	0	0	$\frac{3}{32}$
1	0	1	$\frac{1}{32}$
1	1	0	$\frac{4}{32}$
1	1	1	$\frac{8}{32}$

$$P_i(A=0) P_i(B=0) = \frac{16}{32} \cdot \frac{8}{32} = \frac{4}{32} = P_i(A=0 \cap B=0) \quad \checkmark$$

$$P_i(A=0) P_i(B=1) = \frac{16}{32} \cdot \frac{24}{32} = \frac{12}{32} = P_i(A=0 \cap B=1) \quad \checkmark$$

$$P_i(A=1) P_i(B=0) = \frac{16}{32} \cdot \frac{8}{32} = \frac{4}{32} = P_i(A=1 \cap B=0) \quad \checkmark$$

$$P_i(A=1) P_i(B=1) = \frac{16}{32} \cdot \frac{24}{32} = \frac{12}{32} = P_i(A=1 \cap B=1) \quad \checkmark$$

yes.

is $A \perp\!\!\!\perp B$?

A	B	C	P_2
0	0	0	$\frac{1}{32}$
0	0	1	$\frac{3}{32}$
0	1	0	$\frac{6}{32}$
0	1	1	$\frac{6}{32}$
1	0	0	$\frac{3}{32}$
1	0	1	$\frac{3}{32}$
1	1	0	$\frac{2}{32}$
1	1	1	$\frac{8}{32}$

$$P_2(A=0) P_2(B=0) = \frac{16}{32} \neq \frac{10}{32} = P_2(A=0 \cap B=0) \quad \times$$

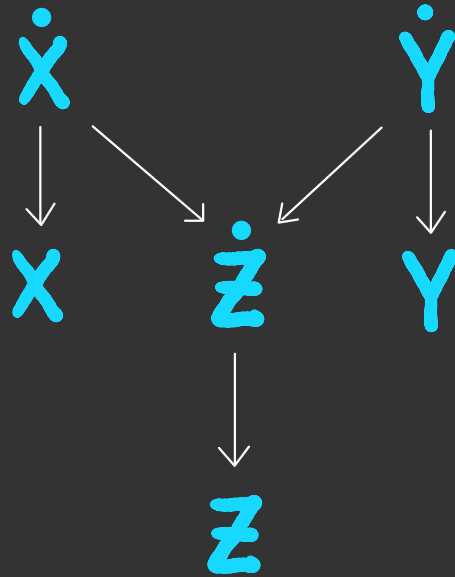
$$P_2(A=0) P_2(B=1) = \frac{16}{32} \neq \frac{12}{32} = P_2(A=0 \cap B=1) \quad \times$$

$$P_2(A=1) P_2(B=0) = \frac{16}{32} \neq \frac{6}{32} = P_2(A=1 \cap B=0) \quad \times$$

$$P_2(A=1) P_2(B=1) = \frac{16}{32} \neq \frac{10}{32} = P_2(A=1 \cap B=1) \quad \times$$

no.

a bayesian
network allows
you to "see"
independence
relationships





if you know
what
to look for

there are only three ways that a **variable** can be connected to **two other variables**

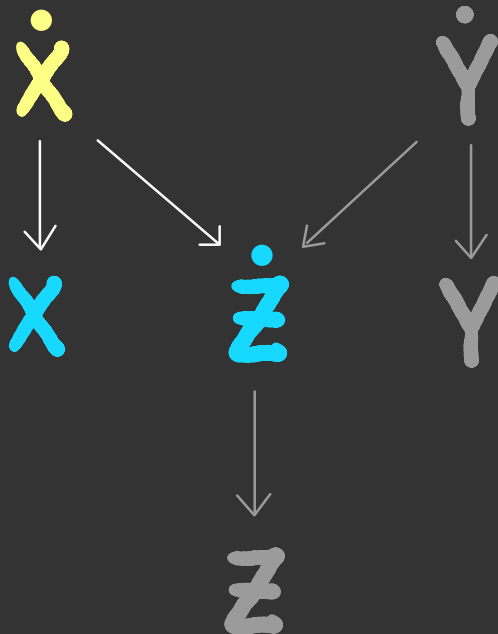
a fork



a chain



a collider



there are only three ways that a **variable** can be connected to **two other variables**

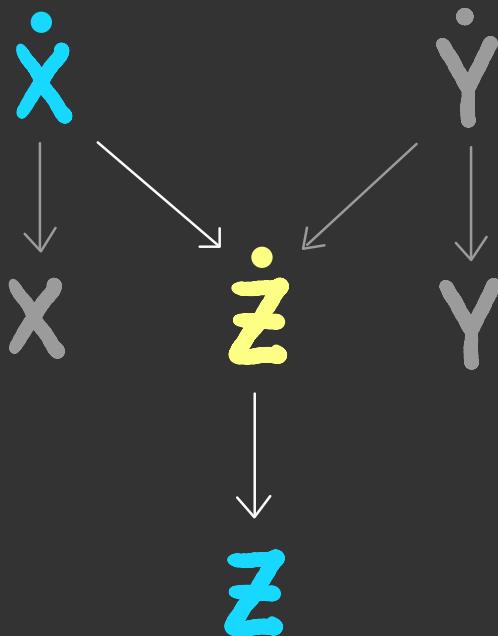
a fork



a chain



a collider



there are only three ways that a **variable** can be connected to **two other variables**

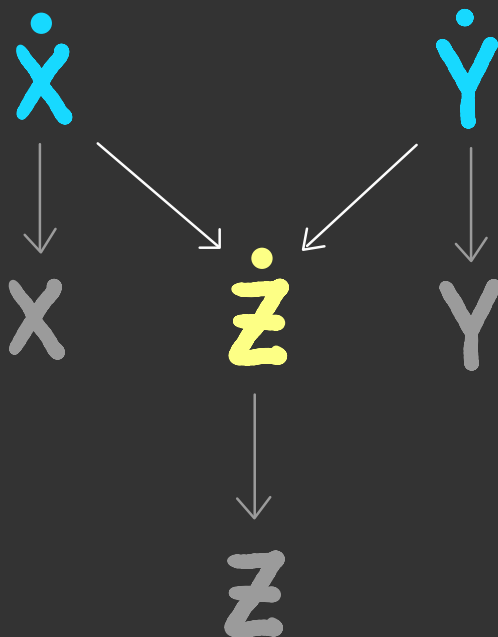
a fork



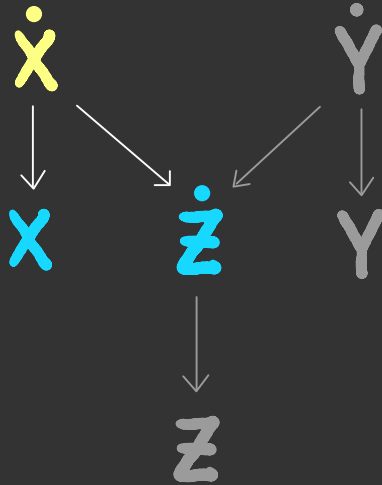
a chain



a collider



a fork



if we **know** $\dot{X}$ (Xena's genotype):

then X and $\dot{Z}$ are
independent

if we **don't know** $\dot{X}$ (Xena's genotype):

then X and $\dot{Z}$ are
not independent

fork



if we know **Z**,
then **X** and **Y** are
independent

$$P(x|y,z) \stackrel{y|x}{=} \frac{P(x,y,z)}{P(y,z)}$$

$$\stackrel{\text{total}}{=} \frac{P(x,y,z)}{\sum_{x'} P(x',y,z)}$$

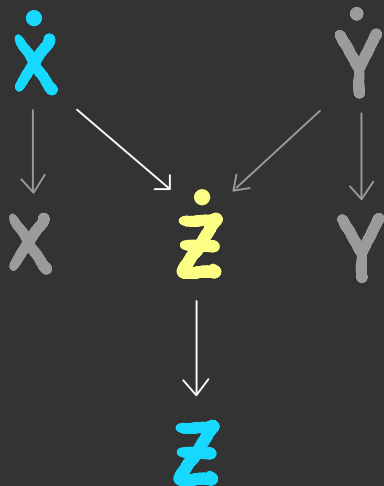
$$\stackrel{x \leftarrow z \rightarrow y}{=} \frac{P(z)P(x|z)P(y|z)}{\sum_{x'} P(z)P(x'|z)P(y|z)}$$

$$\stackrel{\Sigma \rightarrow}{=} \frac{P(z)P(x|z)P(y|z)}{P(z)P(y|z)\sum_{x'} P(x'|z)}$$

$$\stackrel{k/k}{=} \frac{P(x|z)}{\sum_{x'} P(x'|z)}$$

$$\stackrel{\sum p(\cdot) = 1}{=} P(x|z)$$

a chain



if we know $\dot{Z}$ (Zelda's genotype):

then X and Z are
independent

if we don't know $\dot{Z}$ (Zelda's genotype):

then X and Z are
not independent

chain



if we know Z ,
then X and Y are
independent

$$P(X|Y, Z) \stackrel{y|x}{=} \frac{P(X, Y, Z)}{P(Y, Z)}$$

$$\stackrel{\text{total}}{=} \frac{P(X, Y, Z)}{\sum_{x'} P(x', Y, Z)}$$

$$\stackrel{\begin{smallmatrix} \square \rightarrow \square \rightarrow \square \\ x \quad z \quad y \end{smallmatrix}}{=} \frac{P(X)P(Z|X)P(Y|Z)}{\sum_{x'} P(x')P(Z|x')P(Y|Z)}$$

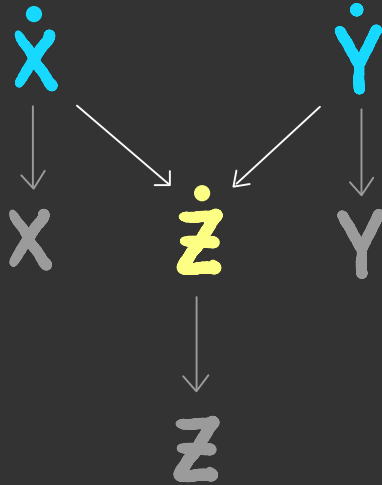
$$\stackrel{\Sigma \rightarrow}{=} \frac{P(X)P(Z|X)P(Y|Z)}{P(Y|Z) \sum_{x'} P(x')P(Z|x')}$$

$$\stackrel{\cancel{k}/\cancel{k}}{=} \frac{P(X)P(Z|X)}{\sum_{x'} P(x')P(Z|x')}$$

$$\stackrel{\text{total}}{=} \frac{P(X)P(Z|X)}{P(Z)}$$

$$\stackrel{\text{man}}{=} P(X|Z)$$

a collider



if we **don't know** $\dot{Z}$ (Zelda's genotype):

then $\dot{X}$ and $\dot{Y}$ are
independent

if we **know** $\dot{Z}$ (Zelda's genotype):

then $\dot{X}$ and $\dot{Y}$ are
not independent

collider



x z y

if we **don't know** z ,
then x and y are
independent

$$P(x|y) \stackrel{\text{total}}{=} \sum_z P(x, z|y)$$

$$\stackrel{y|x}{=} \frac{\sum_z P(x, y, z)}{P(y)}$$

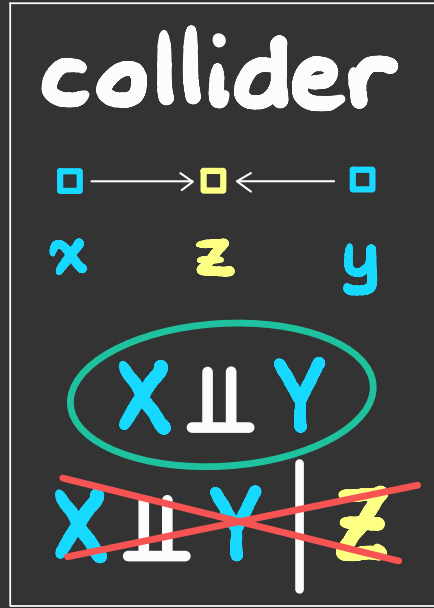
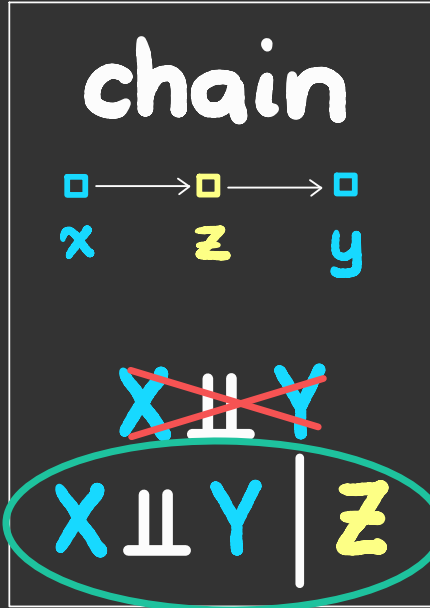
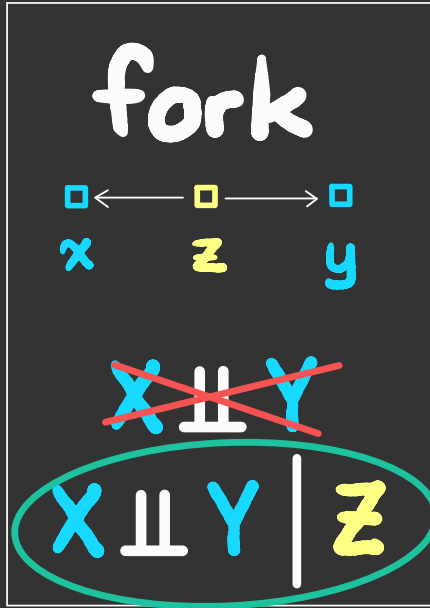
$$\stackrel{\begin{smallmatrix} \square \rightarrow \square \leftarrow \square \\ x \quad z \quad y \end{smallmatrix}}{=} \frac{\sum_z P(x)P(y)P(z|x, y)}{P(y)}$$

$$\stackrel{\sum \rightarrow}{=} \frac{P(x)P(y) \sum_z P(z|x, y)}{P(y)}$$

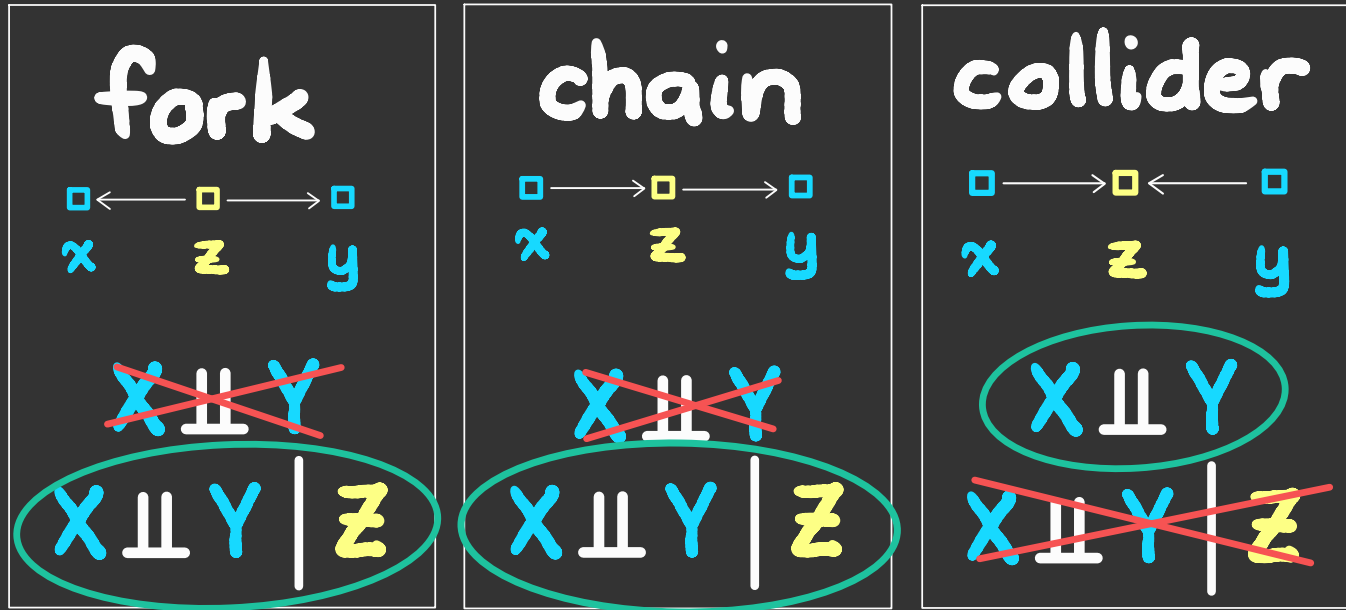
$$\stackrel{\begin{smallmatrix} \cancel{x} \\ \cancel{y} \end{smallmatrix}}{=} P(x) \sum_z P(z|x, y)$$

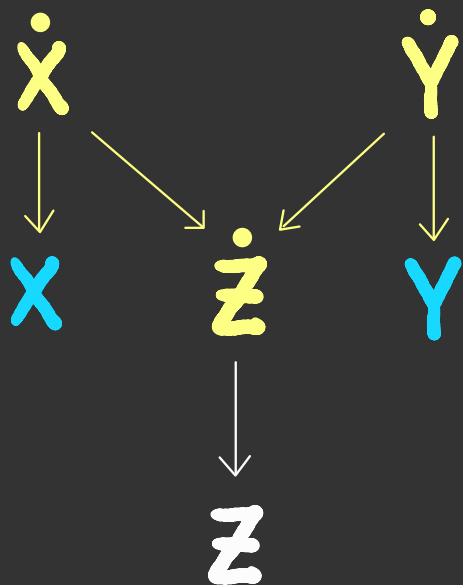
$$\stackrel{\sum P(\bullet) = 1}{=} P(x)$$

summary



we can use these building blocks to determine the flow of relevance in larger networks



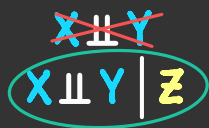


to determine the flow of relevance between two variables, examine every path between them (ignoring direction)

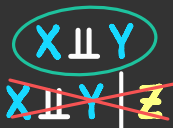
fork



chain

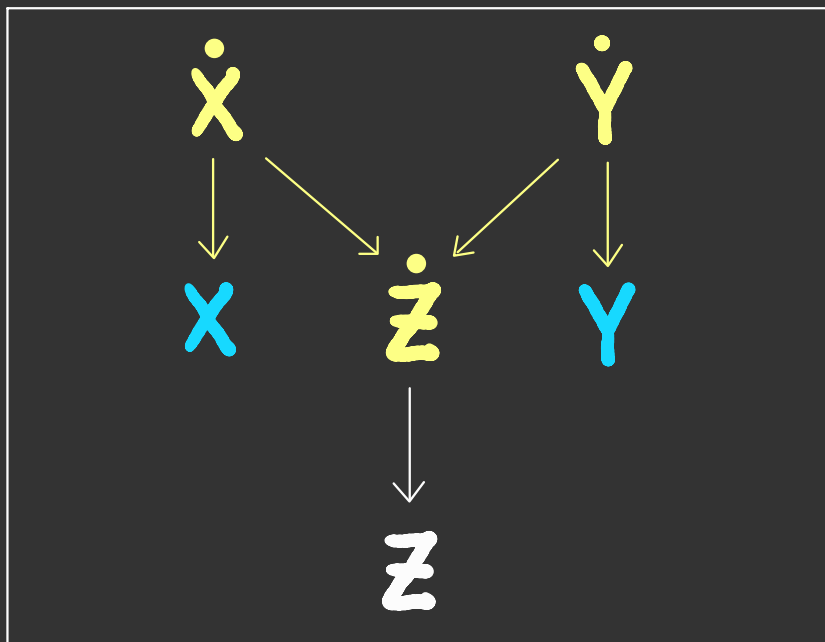


collider

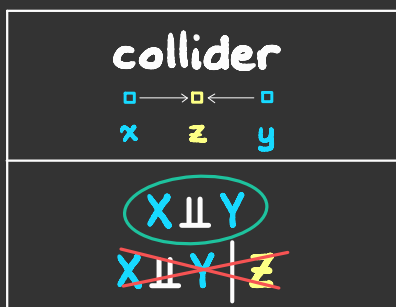
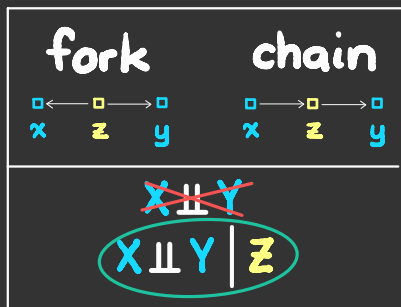


e.g. only path between X and Y:





each intermediate node on a path is the center of a fork, chain, or collider



e.g. only path between X and Y :



fork collider fork

a set W of vertices "blocks" a path if

W contains the center of some

fork or chain



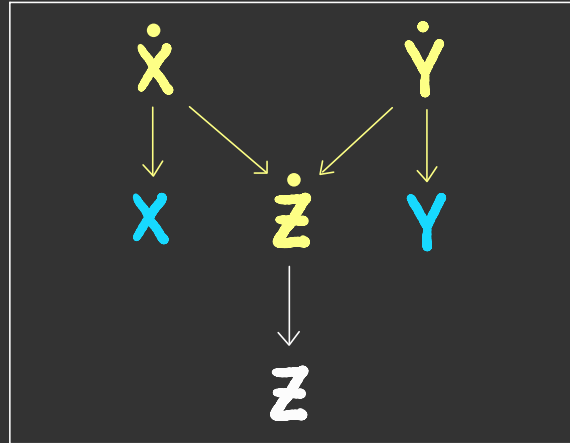
or

W does not contain the center of some

collider



along the path, nor any descendant of the center



a set W of vertices "blocks" a path if

W contains the center of some

fork or chain



or

W does not contain the center of some

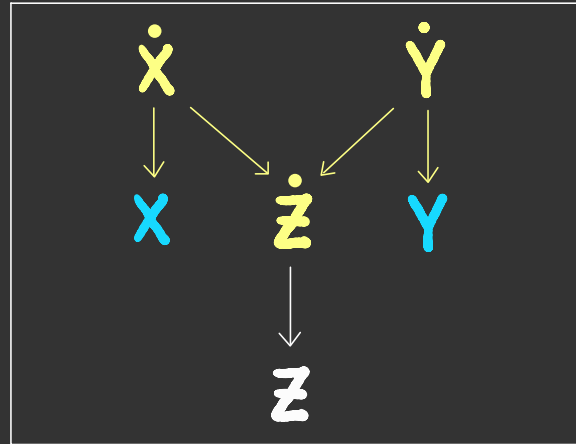
collider



along the path, nor any descendant of the center

e.g.

if W contains $\dot{X}$ or $\dot{Y}$, then W blocks the path from X to Y



a set W of vertices "blocks" a path if

W contains the center of some

fork or chain



or

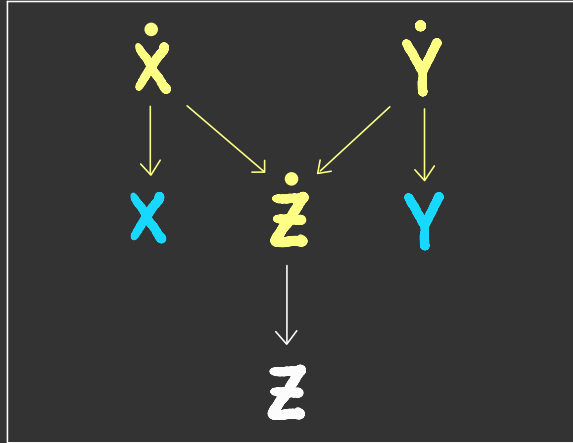
W does not contain the center of some

collider



along the path, nor any
descendant of the center

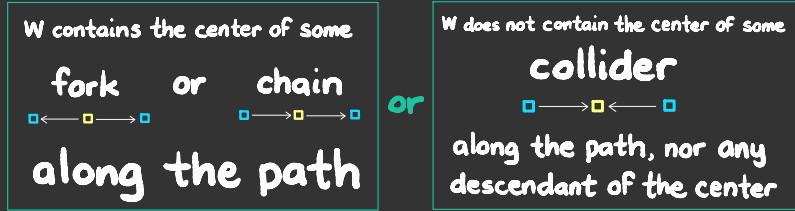
along the path



e.g.

if W contains neither $\dot{Z}$ nor Z ,
then W blocks the path from X to Y

a set W of vertices "blocks" a path if

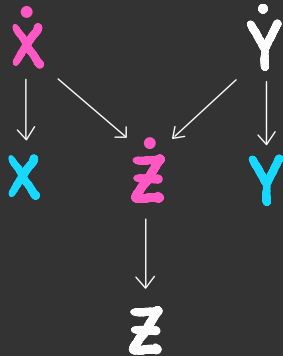


if a set W of vertices blocks every path
between vertices X and Y , we say that
 X and Y are **d-separated** by W
(notation: $X \perp Y \mid W$)

$$X \perp Y \mid W \Rightarrow X \perp\!\!\!\perp Y \mid W$$

"if X and Y are **d-separated** by W in a bayesian network, then for every distribution that factors according to the bayesian network, X and Y are **conditionally independent** given W "

e.g. $X \perp Y \mid \{\dot{X}, \dot{Z}\} \Rightarrow X \perp\!\!\!\perp Y \mid \{\dot{X}, \dot{Z}\}$



for every P such that

$$P(\dot{x}, \dot{y}, \dot{z}, x, y, z) = P(\dot{x})P(\dot{y})P(\dot{z} \mid \dot{x}, \dot{y})P(x \mid \dot{x})P(y \mid \dot{y})P(z \mid \dot{z})$$